

Subject Name: DOWNSTREAM PROCESSING

Subject No: BT30204

Prerequisite: Biochemical Reaction Engineering and Bioenergetics (BT20207)

Objective: This course provides an overview of the unit operations required for product recovery, purification and polishing after fermentation. The course aims to provide an overall idea about processes like centrifugation, filtration, adsorption, chromatographic techniques, crystallization and drying.

LTP- 3-0-0

Credit- 3

Content:

<u>Topic</u>	<u>Hour</u>
Introduction to unit operations in Downstream Processing including cell disruption techniques	3
Purification of proteins by precipitation	2
Solvent extraction of biomolecules	2
Aqueous two phase system for protein purification	2
Reverse Micellar based Separation of Proteins	2
Foam fractionation for protein purification	1
Principles of chromatography in protein purification	1
The Theoretical Plate Model of Chromatography and rate theory of chromatography	2
Gel permeation chromatography	2
Ion exchange Chromatography	2

Affinity chromatography	2
Electrophoresis for protein purification	1
Theories and applications of centrifugation	2
Theory of filtration and filtration equipment	3
Ultrafiltration and tangential flow filtration	1
Introduction to adsorption chromatography	1
Concepts of elution chromatography	2
Concept of yield and purity in elution chromatography	1
Theory of crystallization	1
Continuous Crystallization (MSMPR)	2
Drying: Spray drying and Lyophilization	2
Total	37

Text Books:

1. Bioseparation Engineering by M. Ladisch
2. Bioprocess Engineering by Shuler and Kargi
3. Bioprocess Engineering Principles by Doran